SO cross-cl pipeline: decision tree — Bayesian search + profile scheme

left: search for moments with the evidence (r marginalised) · right: profile scheme, every model at its own r

A · Bayesian search for moments (evidence, r free)

run BBPower without moments — $M_{0,r}$

is χ^2 good? (PTE > 0.01)

No → **Res_wr**: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families, each with its own β free

moment detected? (*) $\Delta\chi^2$ of the residual fit: p < 0.32 (1σ)

Yes → $M_{1,r}$ = $M_{0,r}$ + a detected family, in order of significance (**); full fit (all parameters free), pruned (***)

is $M_{1,r}$ preferred over $M_{0,r}$? (ln B > 0) (****)

Yes → $M_{0,r} = M_{1,r}$: start again

No → $M_{1,r}$ = $M_{1,0}$ with r free, pruned with r free (***)

Res_r0: residuals of $M_{0,0}$ (r = 0) all parameters fixed, own β free; test the families d, s, dT, d2, s2, ds1, ds2

moment detected? (*) $\Delta\chi^2$ of the residual fit: p < 0.32 (1σ)

Yes → $M_{1,0}$ = M_1 + the detected family with the highest significance (**), r = 0; new spectra pruned with r = 0 (***) (none left → the next family)

M_1 = $M_{0,r}$ at the first round, then the $M_{1,0}$ of the previous one

is $M_{1,r}$ preferred over $M_{0,r}$? (ln B > 0) (****)

Yes → $M_{0,r} = M_{1,r}$: start again

No → is $M_{1,0}$ (r = 0) preferred over $M_{0,r}$? (ln B > 0) (****)

No → **Res_r0** on the residuals of $M_{1,0}$ ($M_1 = M_{1,0}$): only the families not discarded before, next round → moment detected? (*)

Yes → → M_1 to the profile scheme

is r of $M_{0,r}$ compatible with zero at 3σ?

Yes → *upper limit*

No → *r detection*

Bayesian result: always the posterior of r of $M_{0,r}$

then the profile scheme (with $M_{0,r}$ and M_1)

is r compatible with zero at 1σ?

Yes → *upper limit*

No → **Res_wr**: residuals of $M_{0,r}$ all parameters fixed at their best fit (r too); test the d and s families, each with its own β free

B · Profile scheme: every model at its own r

M* = $M_{0,r}$, the final model of the tree
candidates c = M* + every combination of the families added in the last Res_r0 (M_1 included), respecting their requirements (ds1 only with s)

$M_{0,r}$ and M_1: the lower and the upper end of the moments; the same candidates whether $M_{1,0}$ was preferred or not

profiles in r
M* → r* = its minimum
every candidate c → r^*_c = its minimum

profile: r fixed on a grid, all the other parameters re-fitted (ML)

compare all the models, each at its own r
M* at r*, every candidate c at r^*_c: local evidence (****) of each (its moment spectra free; β_d, β_s free in all, T_d only with dT)

final model = the one with the highest evidence
(M* itself if no candidate beats it)

profile of the final model in r
r_min = its minimum (r* or r^*_c); posterior of r = exp(-$\Delta\chi^2$/2) × prior of the tree

other parameters
posterior conditional on r = r_min (emcee, r fixed, the rest free)

0 within the central 3σ interval of the posterior of r?

Yes → *upper limit* — r_95 of the posterior with r ≥ 0

No → *r detection* — median and 68% of the posterior of r

(*) ML fit of the family on the residuals (everything else fixed): $\Delta\chi^2$ vs the base fit, χ^2 with k = amplitudes of the family degrees of freedom; detected if p < 0.32 (1σ, the analogue of 0 outside the 68% HPD); the family enters whole

(**) significance: p of the residual test of each detected family; it sets the order in which the families are tried (Res_wr) or picks the one added (Res_r0)

(***) amplitudes with 0 in their 68% HPD (posterior of the full fit) removed and the model refitted, until all are detected; the pruned model is kept if ln B(pruned vs full) > 0 and is the one compared with $M_{0,r}$: for $M_{1,0}$ only the new spectra are pruned, with r = 0

(****) ln B within 2σ of the threshold (A) or between the two best models (B): PolyChord repeated with other seeds, mean of ln Z, up to 3 runs per fit